

Artwork & Plate Verification System — Product Specification

Working name: (TBD — see §14)

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Date: 16 August 2026

Platforms: Flutter (Android APK + Web). Multi-tenant
SaaS — sellable to any printer.

Status: Specification. Pre-build.

1. The Problem (validated with a real job)

Test case: **CGM2026-27-1326 — DAILY KALONJI
100ML LABEL**, Daily Pharma, printed by Prakruti
Graphics.

Three documents exist for this one job, in three
different places:

Document	Where it lives	What it says
Client-approved artwork + job sheet	Email attachment, 17-Apr-26	6 COL + VARNISH , colours: Yellow, Magenta, Cyan, Black, Gold, P 7483 C. Label 114×76mm. Chromo. Flexo.
Vendor separation PDF	Email attachment, 20-Apr-26	6 pages: Cyan, Magenta, Yellow, Black, 617, 7483
Vendor composite proof	Same email	Header: Col: 6

The error: approval sheet declares 7 units (6 colours + varnish). Separation set contains 6. **The varnish plate is absent from the set.**

Why the human eye misses it: the declaration and the plate set are never on screen at the same time. This is not a spelling error, not a design error — it is a **cross-document consistency** failure. No amount of careful reading of any single document catches it.

Second finding from the same job: the mandatory declaration "DAILY PHARMA" (manufacturer name)

exists on **only one plate — P 7483 C**. If that unit misprints, misregisters, or runs dry, the label loses a legally mandatory field while looking otherwise perfect.

Third finding: the job sheet has printed sign-off fields — *Checked by QC Dept, Approved by Pre-Press Incharge, Approved by Manufacturing Incharge*. All three are **blank**. Rev: 01, Date: 00-00-00 . The process is designed but not enforced, because paper cannot enforce anything.

Also note the vendor's own disclaimer printed on their proof: *"...shall not be responsible beyond that, Customers are advised to check plates carefully before printing."* Liability is explicitly transferred to the printer, who has no systematic tool to discharge it.

The core insight

This product is **not** a spell checker. Spell checking is one small module. The product is a **cross-document consistency engine** that holds the job spec, the approved artwork, the separations, and the printed sample in one place and compares them against each other.

2. Product Scope — Three Steps

STEP 1	STEP 2
STEP 3	
Incoming artwork	→ Separation vs
spec	→ Press-side check
(client → printer)	(vendor →
printer)	(plates on machine)
Content & compliance	Plate integrity
Plate presence	
Before quoting	Before plate
making	Before/during run
Saves: rework cycles	Saves: plate cost
+	Saves: reject run
	machine
time	

Build priority: Step 2 → Step 1 → Step 3. Reasoning in §12.

3. STEP 1 — Incoming Artwork Check

Trigger: client sends artwork. Runs before quoting or before sending to plate vendor.

Input: PDF (preferred, has text layer) or AI/EPS/high-res image.

3.1 Text-layer checks (deterministic)

- Extract all text with position, font, size, colour
- **Spell check** — English + Indian languages (Devanagari, Gujarati, Tamil, Telugu, Kannada, Bengali, Marathi). Custom dictionary per customer (brand names, product names, ingredient terms)
- **Number sanity** — decimal placement, unit consistency (mg/g/ml), nutritional table arithmetic where derivable
- **Font size floor** — Legal Metrology prescribes minimum heights for declarations. Flag any mandatory field below the threshold in the active rule pack
- **Duplicate / contradictory fields** — two different MRPs, two net quantities, mismatched batch formats

3.2 Compliance checks (rule-pack driven — see §7)

Applies the rule pack selected for the job's product category and jurisdiction.

Example — FSSAI packaged food pack, applied to the Kalonji label:

Requirement	Found	Status
Manufacturer name + full	Daily Pharma, 111 First Floor, Parvati	✓

Requirement	Found	Status
address	Industrial Premises, Lower Parel, Mumbai- 400013	
FSSAI licence number	11522004000066	✓
Net quantity	100 ml	✓
MRP field with "incl. of all taxes"	present, blank (variable data)	✓ field present
Best before / shelf life	24 months from packing	✓
Veg / non-veg mark	green square symbol	✓
Customer care contact	7977201178 + email	✓
Batch number field	present, blank (variable data)	✓ field present
Country of origin	PRODUCT OF INDIA	✓
Ingredients list	present	✓

Requirement	Found	Status
Nutritional information	present	✓

3.3 Variable-data field handling

Batch No., Pkd., MRP are blank boxes filled at print time. The system must:

- Recognise them as **declared variable fields**, not missing content
- Let the user mark regions as variable during job setup
- Exclude them from "missing text" errors but **include them in field-presence checks**

3.4 Barcode / QR checks

See §8.

3.5 Output

A **query list** the printer sends back to the client — not a pass/fail. Each item: what was checked, what was found, what needs confirmation.

4. STEP 2 — Separation vs Approved Spec (CORE MODULE)

This is the highest-value module and the one to build first.

Inputs:

1. Job spec (digitised approval sheet — §4.1)
2. Client-approved composite artwork PDF
3. Vendor separation PDF (multi-page, one page per plate)
4. Vendor composite proof PDF (optional but useful — carries a header line)

4.1 Digitised job sheet

The existing paper job sheet is already a structured spec. It becomes the system's source of truth:

Job ID / SEQ No.	
File name	: DAILY KALONJI 100 ML
Customer	: DAILY PHARMA
Date	: 17/04/2026
No. of colours	: 6 COL + VARNISH
Colour list	: Yellow, Magenta, Cyan, Black, Gold, P 7483 C
Paper	: Chromo
Label size	: 114 mm x 76 mm
Print type	: Flexo

```
Revision          : 01
Checked by Pre-Press : (name + timestamp)
Checked by QC Dept  : (name + timestamp)
Approved by Pre-Press Incharge : (name +
timestamp)
Approved by Mfg Incharge      : (name +
timestamp)
```

Everything downstream compares against this record.

4.2 Deterministic checks (machine is 100% certain)

A. Plate count

```
Declared units (incl. varnish/coating) vs
separation page count
```

On the test job: declared 7 (6 col + varnish), found 6
→ **FAIL**. This is the flagship catch.

B. Colour name mapping

Vendor uses production names; the job sheet uses commercial names. A per-customer mapping table resolves them:

Job sheet	Separation header	Mapped
Cyan	Cyan	✓

Job sheet	Separation header	Mapped
Magenta	Magenta	✓
Yellow	Yellow	✓
Black	Black	✓
Gold	617	via mapping table
P 7483 C	7483	via mapping table
Varnish	—	MISSING

Unmapped names are flagged for the user to resolve once; the mapping is then remembered.

C. Text completeness — composite vs merged separations

Extract text from the approved composite. Extract text from every separation page and merge. Compare.

- Text in composite but in no separation → **content will not print**
- Text in a separation but not in composite → **unapproved content added**

D. Per-plate text map + single-plate dependency warning

Build a table showing which text sits on which plate.
Then flag any **mandatory declaration that exists on only one plate**.

On the test job:

- Black: ingredients, nutritional facts, usage, storage, address, Lic. No., 100 ml, PRODUCT OF INDIA
- 617 (Gold): duplicate text set — DAILY KALONJI, table structures, Batch/Pkd/MRP labels
- **7483 (Green): "DAILY PHARMA" only** → single-plate dependency on a mandatory field → **WARN**
- C/M/Y: images only, no text

E. Geometry validation

From the separation header (PLATE : 1.14MM LABEL
SIZE : 76.000 X 114MM CLY: 238.125MM PAPER
SIZE : 169MM):

Check	Formula	Test job
Cylinder repeat is a whole tooth count	$CLY \div 3.175$	$238.125 \div 3.175 = \mathbf{75} \checkmark$
Repeat accommodates label rows	(label height × rows) + gaps ≤ CLY	$(114 \times 2) = 228,$ gap 10.125 $\checkmark$

Check	Formula	Test job
Paper width accommodates label columns	(label width × cols) + trims ≤ paper	(76×2)=152 ≤ 169 ✓
Label size matches job sheet	header vs spec	114×76 ✓
Plate thickness recorded	present	1.14mm ✓

F. Header consistency across pages

Job code, date, and Col: count must be identical on every separation page and on the composite. A mismatch means pages from two different revisions have been mixed — a silent and expensive failure.

G. Version lock

Hash the client-approved file at approval time. Every later comparison runs against that hash. If the vendor worked from a different file, the hash mismatch surfaces it immediately. **This alone prevents the single most expensive error class in the industry: correct-looking plates made from the wrong revision.**

H. Ups / imposition count

Composite shows 4-up (2×2). Verify each separation carries the same repeat count and positions.

4.3 Rule-based flags (machine asks, human answers)

- Same text appearing on two plates — intentional overprint/trap, or duplication error?
- Mandatory declaration on a single spot plate
- Instruction present in the approval email/sheet with no corresponding plate (**varnish**)
- Registration mark presence and consistency across all pages
- Element position drift between plates beyond tolerance

4.4 Out of scope for automation (human judgement)

- Trap and overprint intent
- Colour choice, shade approval, ink sequence
- Aesthetic decisions

The report must state these are not checked, so no one assumes they were.

5. STEP 3 — Press-Side Check

Context constraint that drives the design: the machine is stopped, an operator and QC are waiting, and the whole check must complete in **under ~60**

seconds. A flatbed scanner at the press is not practical. Mobile camera is a **core requirement**, not an optional phase.

5.1 Why this is technically easier than general OCR

The system already knows what the text should be. This is **constrained verification**, not open-ended reading. Expected text acts as a dictionary and expected layout gives positional priors — accuracy is far higher than general-purpose OCR on the same image.

5.2 Scope reduction after Step 2

If Step 2 passed, content is already verified. Step 3 only needs to answer: **are all plates mounted and printing?**

On the test job that means checking:

- Gold elements present (top border, 100ml band, "Kalonji Oil" underline)
- Green "DAILY PHARMA" present
- Varnish window applied
- All four CMYK images present
- Registration marks aligned

This is **spot-check, not full-text OCR** — which is what makes the 60-second budget achievable.

5.3 Capture design

- **Corner alignment guides** on screen — user frames the label. Enables homography/perspective correction so the captured image is warped into the approved artwork's geometry
- **Burst capture** — take 3–5 frames, auto-select sharpest. Invisible to the user
- **Pre-loaded job** — the approved PDF must be selected during machine setup, *before* the press stops. At the press: photo only
- Works on a stopped press with a sample pulled onto the inspection table (flat, reasonable light) — not on a moving web

5.4 Output must never be PASS/FAIL

3 areas need your eyes:

△ Gold plate – "Kalonji Oil" underline not detected

△ MRP zone – low confidence, check manually

△ Green plate – "DAILY PHARMA" faint

47 other elements matched ✓

Reasoning: this converts a 200-word manual read into looking at 3 specific places. It keeps the human as

the decision-maker, which is both operationally correct and the right liability posture (§10).

5.5 Offline requirement

Press floors have poor connectivity. Step 3 must work **fully offline** with the job pre-downloaded, syncing results when connection returns.

6. Additional Modules (beyond the three steps)

These are what turn a tool into a product a printer will pay monthly for.

6.1 Digital sign-off & audit trail — HIGH PRIORITY

Replaces the blank paper sign-off boxes. Each stage records who approved, what file version, timestamp, device. Immutable log.

Why this matters commercially: it converts the vendor's liability disclaimer from a threat into a defence. When a dispute happens, the printer has a record.

6.2 Revision management

Rev 01, 02, 03... with diff between revisions. Answers "what changed since last time?" — currently answered by memory.

6.3 Customer approval portal

Client receives a link, views artwork, approves or comments. Approval is captured with hash + timestamp. Removes the email chain entirely.

6.4 Job library & re-order verification

Repeat orders are common. Store every approved job. On re-order, diff the new file against the last approved version — usually only MRP or batch format changed. **Turns a full review into a 10-second diff.**

6.5 Reports

Exportable PDF per job: what was checked, results, who signed off, file hashes. Attach to the job bag. Send to client. Use in disputes.

6.6 Braille inspection (Phase 3)

Required for pharma exports to the EU. Detection, dot spacing, decoded value. Only relevant if pharma export customers are targeted.

6.7 Font size verification

Legal Metrology prescribes minimum heights for specified declarations. Measure actual rendered height in the PDF against the rule pack threshold.

6.8 Colour / plate ink sequence record

Not colour *approval* (needs spectrophotometer), but recording the declared sequence and flagging deviation from the job sheet.

6.9 Ink coverage estimate (nice-to-have)

Per-plate coverage percentage from separations. Useful for costing — and links naturally to the calculator tools discussed separately.

6.10 Multi-language declaration check

Indian labels often carry declarations in multiple languages. Verify each required language block is present per the rule pack.

7. Compliance Rule Packs — How Norms Actually Work

This is the most important architectural decision in the product.

7.1 The principle

The application does not know any law. It knows how to apply a rule pack.

Rules are **never** hardcoded. They are versioned JSON documents fetched from the server. When a regulation changes, a new rule pack is published — every client gets it immediately, with no app update, no Play Store release, no per-customer deployment.

7.2 Rule pack structure

```
{
  "pack_id": "fssai-packaged-food",
  "version": "3.1",
  "jurisdiction": "IN",
  "category": "packaged_food",
  "effective_from": "2026-04-01",
  "source_reference": "FSSAI Labelling &
Display Regulations",
  "last_verified": "2026-08-01",
  "verified_by": "internal",
  "disclaimer": "Checklist aid. Not legal
certification.",
  "rules": [
    {
      "id": "mfr_name_address",
      "label": "Manufacturer name and
complete address",
      "type": "field_presence",
      "severity": "mandatory",
      "detection": ["regex",
"layout_region", "manual_confirm"]
    }
  ]
}
```

```

    },
    {
      "id": "fssai_licence",
      "label": "FSSAI licence number",
      "type": "pattern",
      "pattern": "^[0-9]{14}$",
      "severity": "mandatory"
    },
    {
      "id": "net_quantity",
      "label": "Net quantity declaration",
      "type": "field_presence",
      "severity": "mandatory",
      "min_font_height_mm": 1.0
    }
  ]
}

```

7.3 Pack library (build progressively)

Pack	Priority	Notes
Legal Metrology — packaged commodities	1st	Applies to nearly every retail pack. Broadest reuse.
FSSAI — packaged food	2nd	Highest volume category
Cosmetics	3rd	

Pack	Priority	Notes
Pharma / drug labelling	4th	Most complex, highest value
Liquor — state-wise	5th	Every state differs. Separate pack per state.
Plastic / EPR marking	6th	
Export destination packs (EU, US, GCC)	Later	

7.4 How a pack is selected for a job

During job setup: product category + jurisdiction (+ state, where relevant). The system loads the matching pack and shows which pack version was used in the report.

7.5 Three honest limits — must be visible in the product

1. **The app is not a compliance authority.** It runs a checklist someone wrote. Report wording: *"11 of 12 checklist items found. 1 not detected."* Never *"COMPLIANT ✓"*.

2. **Rule packs need a maintainer.** Initially you. As the pack library grows this becomes real ongoing work — and eventually a reason to charge more, or to partner with a compliance consultant.
3. **Every pack carries its own provenance** — source reference, last-verified date, verifier. Visible to the user. If a pack is stale, say so.

7.6 Customer-specific packs

Beyond regulation: a brand's own rules ("logo clear space $\geq$ 5mm", "always include #TAGLINE", "MRP font never below 8pt"). Same engine, customer-scoped pack. **This is a strong upsell and a retention lock.**

8. Barcode & QR Verification

Two entirely different problems. Keep them separate in the product.

8.1 In artwork (Step 1 / Step 2) — fully solvable

Extract the barcode from the PDF (vector or embedded raster), then:

- **Decode** the symbol
- **Checksum validation** — EAN-13/UPC check digit must be correct. A wrong check digit means the

barcode will not scan, and this is entirely deterministic

- **GS1 structure parsing** — GTIN, and Application Identifiers for batch (10), expiry (17), serial (21) where present
- **Value match** — decoded GTIN vs the value recorded in the job spec. Catches the case where artwork carries a previous product's barcode
- **Quiet zone** — required clear space present
- **X-dimension / magnification** — within the standard's permitted range for the symbology
- **Bar Width Reduction** — flexo requires BWR compensation. Record what the job spec declares and whether it was applied
- **Colour combination** — bars must be dark on light with adequate contrast. Flag red/light bars, or bars printed on a spot colour with poor contrast
- **QR** — decode, verify payload matches spec, check error-correction level and module size

8.2 On printed sample (Step 3) — partial only, be honest

- **Decode / no-decode** — a phone camera can reliably tell you whether the printed barcode scans at all. This is genuinely useful and catches the worst failure

- **Formal ISO/IEC 15416 grading (A-F) — not possible with a phone camera.** Grading requires a calibrated verifier with controlled illumination and aperture

Product rule: never display a grade from a phone capture. Show "scanned successfully" or "failed to scan", and state that formal grading requires a verifier. Claiming a grade you cannot measure destroys credibility with exactly the customers who know the difference.

9. Technical Architecture

9.1 Stack

Layer	Choice	Reasoning
Mobile	Flutter (Android APK)	Press floor, offline, camera
Web	Flutter Web	Pre-press desk, large PDFs, dual- monitor review
Auth / DB / Storage	Firebase (Auth,	Matches existing stack

Layer	Choice	Reasoning
	Firestore, Storage)	
PDF/CV processing	Python service on Cloud Run	See 9.2
Rule engine	JSON packs + evaluator (server-side)	Hot-updatable
Offline	Local cache + sync queue	Press floor connectivity

9.2 Why processing cannot live in Flutter or Cloud Functions

PDF text extraction with position data, page rendering, image diffing, homography, and barcode decoding need mature native libraries. Cloud Functions are too constrained on memory, execution time, and binary dependencies for multi-page separation PDFs.

Run a Python service on Cloud Run:

- `PyMuPDF` — text with coordinates, font metrics, page rendering
- `pdfplumber` — table and layout structure
- `OpenCV` — image diff, alignment, homography

- `pyzbar` / `zxing-cpp` — barcode decode
- `Pillow` , `NumPy` — image handling
- OCR: `PaddleOCR` or `Tesseract` (constrained mode with expected-text dictionary)

Flutter handles UI, capture, and job workflow. All heavy work is an API call.

9.3 Multi-tenancy (required — this is sold to any printer)

- Tenant = printer company
- Per-tenant: users, roles, customers, jobs, colour mapping tables, custom dictionaries, custom rule packs, branding on reports
- **Strict data isolation.** Printers compete with each other; a leak ends the business
- Shared: regulatory rule packs, symbology standards, engine

9.4 Roles

Role	Can do
Admin	Everything, user management, billing
Pre-press	Create jobs, upload artwork, run Steps 1 & 2

Role	Can do
QC	Run all checks, sign off
Press operator	Step 3 only, view assigned jobs
Manager	Approve, view all reports, dashboards
Customer (external)	View own jobs, approve/comment only

9.5 Data handling — a hard requirement

Printers will not upload customer artwork to a system they do not trust. Non-negotiable:

- Explicit retention policy, stated in plain language
- Per-tenant encryption at rest
- Configurable auto-delete after job completion
- **On-premise deployment option** for large customers — this will be asked for
- No cross-tenant training or analytics on artwork content, ever

10. Liability Positioning

The vendor already prints *"Customers are advised to check plates carefully before printing."* This product

must not absorb that liability.

Rules baked into the product:

1. Never output "APPROVED", "PASS", or "COMPLIANT". Output findings and counts.
2. Every report states what was **not** checked (colour, trap intent, aesthetics).
3. Every report names the human who signed off.
4. Rule pack version and last-verified date appear on every compliance report.
5. Terms of service state clearly: decision-support tool, not a guarantee.

Framed correctly, this is a **selling point**, not a weakness: the product gives the printer a documented, timestamped record of diligence — which is exactly what they currently lack when a dispute arises.

11. Commercial Model

This is a paid product from day one. It is not a free traffic tool.

Value anchor: one rejected label run costs lakhs in material, machine time, and customer relationship. One plate set remade costs plate charges plus

schedule slip. Against that, ₹2,000–5,000/month is trivial.

Tier	For	Approx
Starter	1 user, Step 1 + 2, limited jobs/month	₹1,500–2,500/mo
Standard	5 users, all 3 steps, audit trail, reports	₹4,000–6,000/mo
Pro	Unlimited users, customer portal, custom rule packs, API	₹10,000–15,000/mo
Enterprise	On-premise, integrations, SLA	Custom

Adjacent revenue: custom rule pack authoring, onboarding/training, and integration with the existing CRM/ERP work (Flexora) — same customer, same conversation.

12. Build Order & Milestones

Phase 0 — Validation (2–3 weeks)

Script-only, no UI. Run Step 2 checks over **the last 20 jobs at PGPL**.

- Reprocess the Kalonji job — the varnish gap must be caught. **This is regression test #1**
- Count how many other real errors surface
- **Decision gate:** if 20 historical jobs surface fewer than 2–3 genuine issues, reconsider scope before building further

Phase 1 — Step 2, web only (6–8 weeks)

Job spec entry, three-file upload, all deterministic checks, findings report, PDF export.

Success: PGPL uses it on every job for one month without falling back to manual.

Phase 2 — Step 1 + sign-off (6–8 weeks)

Spell check, variable-field marking, first two rule packs (Legal Metrology, FSSAI), barcode artwork checks, digital sign-off, audit trail, revision management.

Success: 3 external printers paying.

Phase 3 — Step 3, Flutter APK (8–10 weeks)

Camera capture with alignment guides, offline job cache, spot-check comparison, sync.

Success: used at the press on live jobs, under 60 seconds per check.

Phase 4 — Scale

Customer approval portal, job library re-order diff, more rule packs, on-premise option, integrations.

Why Step 2 first: cheapest to build (pure PDF, no camera, no OCR), highest saving per catch (before plates are made), and there is already a proven failure case to test against.

13. Risk Register

Risk	Severity	Mitigation
Separation PDFs vary by vendor (headers, naming, structure)	High	Build a parser profile per vendor. Collect samples from 5+ plate makers before finalising the parser
Printers won't upload customer artwork	High	On-premise option, clear retention policy, auto-delete
Rule packs go stale	High	Provenance on every pack, visible last- verified date,

Risk	Severity	Mitigation
		quarterly review cycle
False positives cause distrust	High	Never PASS/FAIL. Show confidence. Let users dismiss and remember dismissals
Step 3 too slow for press reality	Medium	Pre-load job before the press stops; spot-check not full OCR
Multi-tenant data leak	Critical	Strict isolation, per-tenant encryption, audit logging
Enterprise incumbent moves down-market	Medium	Existing tools are explicitly built for large organisations. Price and simplicity are the moat
Scope creep across three steps at once	High	Ship Step 2 alone. Resist the rest until it is in daily use

14. Open Items

- [] Product name and domain (must not be tied to any one step)
 - [] Confirm what actually happened with the varnish plate on the Kalonji job — separate later file, or genuinely missed?
 - [] Collect separation PDFs from 5+ different plate vendors to test parser robustness
 - [] Verify all Legal Metrology minimum font heights against the current source text before encoding them
 - [] Decide: is the customer-facing approval portal in scope for v1, or Phase 4?
 - [] Pricing validation with 5 printers before publishing tiers
 - [] Confirm colour mapping conventions (Gold → 617, P 7483 C → 7483) — is this vendor-specific or an industry convention?
-

15. One-Line Summary

A cross-document consistency engine for label and packaging jobs: it holds the job spec, the approved artwork, the plate separations, and the printed sample together and tells a human exactly

**where to look — with an audit trail proving they
looked.**